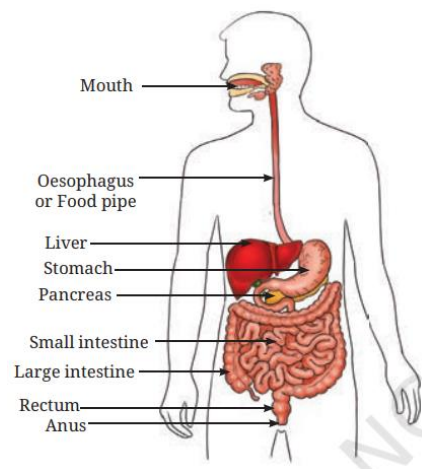


Chapter – 9: Life Processes in Animals

If your food is fully digested before you eat again, you won't need medicine for pain. – Thirukkural 942

- Grade 6 – textbook – *curiosity* – chapter – living creatures – processes imp. – survival – nutrition, respiration, excretion, reproduction
- This chapter – learn – life processes – nutrition, respiration – detail
- **Observe** – surroundings – **notice** – what animals eat
- Animals – diff. types – food
- Bees, sunbirds – suck nectar – infants – humans, other animals – mother's milk
- Snakes – swallow animals – aquatic animals – filter food particles – floating nearby – feed on them
- Animals – humans – energy – food – carry out – life processes
- Animals – consume food – complex components – carbs, proteins, fats
- These complex food – need to break down – simpler forms – before body uses them
- Breaking down – complex food – simpler forms – long tube – **alimentary canal**
- This process – start – mouth – end – anus
- Food – move through canal – digestive juices – diff. parts – break it – simpler forms
- This form – absorbed – diff. parts – alimentary canal – transport – various parts – body – various func.



Nutrition in Animals

- How – complex food – breaks down – simpler forms – various animals?
- This process – same – all animals – OR – vary?

Digestion in human beings

- Trace journey – food – inside body – pass through – diff. parts – alimentary canal
- **Beginning with mouth cavity**
 - Journey – food – begins – enter mouth
 - Teeth – break down food – smaller pieces – crushing, chewing
 - This process – initial breakdown – food – fine pieces – **mechanical digestion**
 - Think – favorite food – mouth watery – reason – **saliva** – released
 - Take – small piece – *chapati* – OR – boiled rice – bite size portion – chew properly – 30-60 seconds
 - 1st – *chapati*, rice – usual taste – continue chewing – notice change – tastes sweet
 - *Chapati*, rice – contain starch – carb
 - Saliva – digestive juice – help – break down – starch to sugar
 - Starchy food – taste sweet – chew – longer
 - Activity –
 - Take – 2 test tubes – label – A, B
 - Test tube A – 1 teaspoon boiled rice – test tube B – 1 teaspoon chewed boiled rice
 - Add – 3-4 mL water – both test tubes
 - Note – initial colour – rice-water mixture
 - Add 3-4 drops – iodine solu. – each test tube – dropper
 - Mix – content – each test tube – separately – obs.

- Obs. – colour – test tube A – blue-black – test tube B – no / min. change
- Grade 6 – learnt – iodine – reacts with starch – blue-black colour
- Test tube A – blue-black colour – **indicate** – starch presence
- Test tube B – no blue-black colour – indicate – no starch
- Starch – converted to sugar – act of saliva
- We know – saliva secretion – helps break down – starch to sugars
- This process – complex food to simpler forms – digestion
- Food – partially digested – mouth – further digestion – alimentary canal
- **Food pipe (Oesophagus): A passage from mouth to stomach**
 - Chew food – saliva – helps digesting starch – also – moistens – soft, easy – swallow
 - Tongue – help – mixing food with saliva – pushing – long, flexible tube – food pipe – **oesophagus**
 - Walls – food pipe – contract, relax – wave motion – push food – stomach
 - This movement – throughout alimentary canal – push food
- **Stomach**
 - Walls – contract, relax – churn food
 - Churned food – mixed – secretion – inner lining – stomach
 - Stomach secretions – digestive juice, acid, mucus
 - Digestive juice – stomach – breaks down – proteins – simpler components
 - Acid – help break protein – ALSO – kill harmful bacteria
 - Mucus – protect stomach lining – prevent damage
 - Stomach – food – partial digestion – convert to – semi-liquid mass – prepare it – next stage
- **Small intestine**
 - After stomach – partially digested food – move – small intestine
 - Small intestine – length – surprise – 6 m – almost 2x height – classroom
 - This part – longest – alimentary canal
 - Small intestine – digestive secretions – 3 src. –
 - Inner lining – small intestine
 - Liver
 - Pancreas
 - **Liver** – bile – mildly basic – recall neutralisation – chapter – exploring substance
 - Bile – neutralize acid – break down fats – tiny drops – digestion easy
 - **Pancreas** – pancreatic juice – also basic – help neutralize acid
 - Pancreatic juice – break down – carbs, proteins, fats
 - Intestinal secretions – break down – fats, proteins, carbs – simpler forms
 - Digested nutrients – pass – intestine to blood – blood vessels – walls
 - This process – **absorption**
 - Inner lining – small intestine – thin – 1000s – finger-like projections – increase surface area – absorption
 - These projections – allow – nutrients – pass into blood – carry them – diff. parts – body
 - Nutrients – provide energy – support – growth, development – help body – func. properly
- **Large intestine**
 - Most nutrients – digested, absorbed – small intestine
 - What happens – undigested food? – moves – large intestine
 - Large intestine – length – 1.5 m – shorter than small intestine – called so – much wider
 - This intestine – absorb – water, salts – undigested food – waste – semi-solid
 - This waste – **stool** – stored – lower part – large intestine – **rectum** – until – body – ready to discard

- Fibre-rich food – fruits, vegetables, whole grains – help – large intestine – func. properly – stool pass easier
- Finally – expelled – anus – process – **egestion**

Do all animals digest food same way?

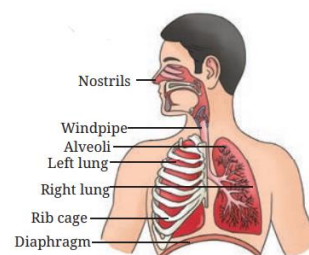
- Grass-eating animals – cows, buffaloes – chew grass – swallow it – stomach
- Stomach – partial digestion – food
- Partially digested food – brought back – mouth – gradual chewing
- This process – rumination – these animals – ruminants
- Cow – 8 hr. daily – chewing food
- Thoroughly chewed food – pass down – alimentary canal – further digestion
- Birds – no teeth – BUT – chamber – gizzard
- Food – breaks down – contraction, relaxation – walls – gizzard – often – aided by – small stones – birds swallow
- Animals – variations – structure, func. – alimentary canal – adapt – diff. ways – digestion – diff. food
- Learnt – nutrients – digestion – carried – diff. parts – body
- Some nutrients – help – build, repair – body – other – sugar – broken down – release energy
- Process – nutrients – converted – usable energy – **respiration**

Respiration in Animals

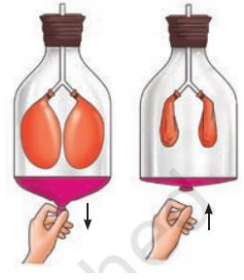
- Learnt – Grade 6 – chapter – living creatures – all living beings – respire
- This process – same – all animals?

Respiration in humans

- You know – people – breathe in (inhale) – breathe out (exhale) – air – obtain O₂ – release CO₂
- Breathing, respiration – diff.?
- **How do we breathe?**
 - Process – inhaling, exhaling air – breathing
 - Difficult to live – without food – 1 week – without water – 1 day – BUT – without breathing – not more than few min.
 - All animals, plants, humans – breathe – stay alive
 - Food – follow – specific path – digestive sys. – our body – specific sys. – breathing, respiration
 - This sys. – **respiratory system** – various parts
 - This sys. – exchange of gases – specific pathway – various parts – assist – breathing, respiration
 - This sys. – start – pair – nasal openings – **nostrils** – inhale, exhale air
 - Inhaled air – pass through – small passages – **nasal passages**
 - Nostrils – tiny hair, mucus – trap – dirt, dust – inhaled air
 - This reason – breathe – nose – not mouth
 - After nasal passages – air – reach lungs – through windpipe
 - Windpipe – 2 branches – enter 2 lungs – in there – branches – divide further – smaller branches – ending – balloon-like sacs – alveoli
 - Lungs – protected by – rib cage
 - Activity –
 - Take – wide transparent plastic bottle with lid – remove bottom
 - Make hole – bottle lid – also take Y-shape hollow tube



- Fix – 2 deflated balloons – forked end – secure them – rubber bands – airtight
- Insert – straight end – lid – open base – seal with clay
- Open base – attach – thin rubber sheet – secure – large rubber band
- Pull – rubber sheet – centre – downwards – obs.
- Release – rubber sheet – upwards – obs. balloons
- Rubber sheet – pulled down – balloons inflate
- Vice-versa – release sheet – upwards – balloons deflate
- Breathe in – chest expands – ribs – move up, outwards – diaphragm – dome-shaped – below lungs – downward movement – inhalation
- This process – increase space – inside chest – air enters lungs
- Breathe out – ribs – move down, inwards – diaphragm – upward movement – exhalation
- This process – decrease space – inside chest – air exits lungs
- This model – balloons – lungs – rubber sheet – diaphragm



• What do we breathe out?

- Activity –
 - Take – equal amt. – fresh lime water – 2 test tubes – A, B
 - Test tube A – pass air – syringe / *pichkari* – same air – inhale
 - Test tube B – blow air – mouth – using straw
 - Obs. – change in colour – lime water
- Lime water – test tube B – turns milky – BUT – test tube A – no change
- Lime water – turns milky – react with CO₂ – indicate – exhaled air – more CO₂ than inhaled air

• How does exchange of gases happen?

- Process – breathing – fresh air – outside – enter lungs – fills alveoli
- Alveoli – thin walls – surrounded – fine tubes – contain blood
- Blood – carry CO₂ – body to alveoli – released into air
- Same time – O₂ – alveoli to blood – transported – all body
- Food – provide energy – how?
- Key – not only food – BUT also – oxygen
- Eat food – body breaks it – simple forms – sugar (glucose)
- Oxygen – helps – break down – glucose – release energy
- This process – **respiration** – word equation –
 - Glucose + Oxygen → Carbon dioxide + Water + energy
- Breathing – inhale air – surroundings – exhale air – more CO₂ – not all O₂ – used
- Some animals – use – larger fraction – respiration
- This exchange – gases – ensure – each segment – body – receive O₂ – produce energy – remove waste
- Breathing – take in O₂ – remove CO₂
- Respiration – use O₂ – break down food – release energy
- This energy – help – walk, run, play, think
- Breathing – physical – respiration – chemical – both – essential
- Our body – unique sys. – transport – nutrients, oxygen, other substances
- This sys. – circulatory sys. – heart, blood, blood vessels – ensure – transport – good things – waste products – removed

Do other animals breathe same way as humans?

- Learnt – diff. animals – diff. habitats – birds fly – fish swim
- Animals – birds, elephants, lions, cows, goats, lizards, snakes – breathe – lungs

- Above animals – lungs – structure – diff.
- Aquatic animals – fish – special structures – **gills** – rich supply – blood vessels
- Exchange – O₂, CO₂ – blood, dissolved gases – across gills
- Amphibians – frogs – live – water, land – use diff. parts – breathing – diff. stages – life
- Ex. – tadpoles – gills – adult frogs – lungs – land – skin – water
- This adaptation – help them survive – both – water, land
- Earthworm – moist skin – exchange of gases
- Diff. animals – diff. apparatus – breathing – suit – their environment
- Apart from digestive sys., respiratory sys., circulatory sys. – other sys. – coordinate – each other – perform – diff. func. – sustain life